

Market power modeling and restraint of aggregated prosumers in peer-to-peer energy trading: A game-theoretic approach

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HIGHLIGHTS

- A framework for P2P energy trading market with the aggregator is established.
- A non-cooperative game model is established for the market competition.
- The impact of market power on trading fairness and efficiency is analyzed.
- The tentative offers penalty is adopted to restrain market power.

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ABSTRACT

The aggregator can be introduced to the community microgrid to facilitate small-scale prosumers participating in peer-to-peer (P2P) energy trading. However, when the aggregator performs price anticipation for profit with its market power, market trading fairness and efficiency of the P2P market may be harmed. To analyze the formation and negative effect of market power, this paper proposes a market power modeling and restraint method of aggregated prosumers with a game-theoretic approach. First, a framework for the P2P energy trading market with the aggregator is established, and the price impact factor is defined and introduced to characterize the impact of the aggregator's trading behavior on the market price. Then, the competition between the aggregator and individual prosumers is modeled as a non-cooperative game, and the impacts of market power on price formation, trading profit, and overall benefit are analyzed, respectively. Finally, the tentative offers penalty is introduced in the market to restrain the market power of the aggregator. Numerical results show that the aggregator can take advantage of its market power to manipulate the market price for profit when the value of the price impact factor is well perceived, which may harm the fair distribution of welfare among prosumers and the overall benefit of the microgrid. Meanwhile, the results also prove that the involvement of the tentative offers penalty can restrict the market power perception ability of the aggregator and further restrain the impact of market power on the benefit of the market.

1. Introduction

Towards the prospect of carbon neutralization, community microgrid is undergoing rapid development to accommodate a higher penetration of renewable generation and distributed energy resources (DERs) [1,2]. Microgrid users will transform from passive consumers to prosumers, which are endowed with the capabilities of energy production, consumption, and storage [3]. The intermittency, uncertainty, and

ownership complexity of the proliferating DERs will bring challenges to the optimal operation of the community microgrid [4]. In such cases, peer-to-peer (P2P) energy trading is proved to be a viable energy management solution for the community microgrid to achieve economic benefit promotion [5], imbalance power elimination [6], renewable energy self-consumption [7] and line loss reduction [8] in many studies.

The involvement of diversified prosumers can effectively promote the competition and flexibility of the P2P market [9]. However, as an

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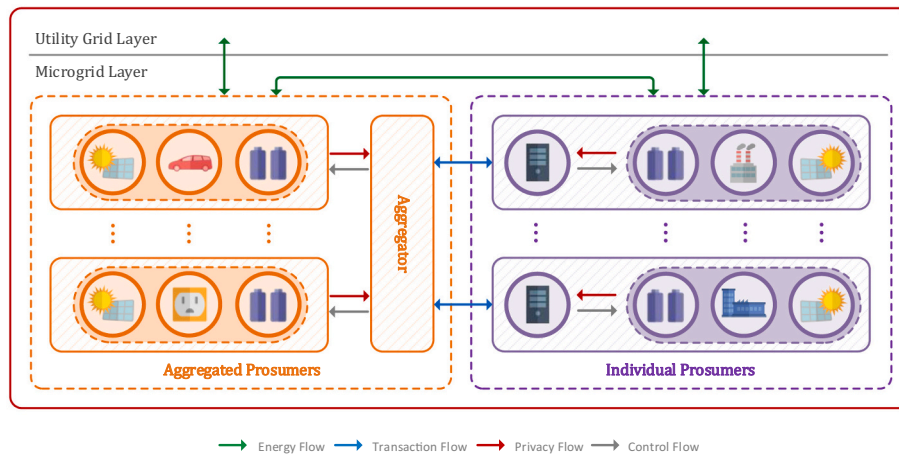


Fig. 1. The structure of the community microgrid peer-to-peer market with the aggregator.

essential role of energy trading, small-scale prosumers (e.g. residential prosumers) may suffer beneficial loss in the competitive market due to limited negotiating ability and bidding flexibility [10], which may further weaken their motivation for energy trading participation. As a solution, the aggregator can be introduced to the community microgrid to facilitate small-scale prosumers participating in the market [11,12].

An aggregator can be regarded as a broker service provider that gathers energy resources of the aggregated prosumers for unified dispatching and market bidding [13]. Endowed with superior scheduling and bidding ability, the aggregator can determine the optimal bidding strategy by utilizing the flexibility potential of energy resources [14] and multiple uncertainties of the market [15] to guarantee the deserved benefit of aggregated prosumers [16]. A careful review of prior works shows that the scheduling and bidding strategy of the aggregator has been the focus of many recent studies [17]. Specifically, a two-stage stochastic optimization model was proposed for the optimal bidding of the aggregator in [18], which effectively reduced the trading cost of aggregated prosumers in day-ahead and real-time market stages. Aiming at reducing the risk of profit loss caused by price volatility, an optimal bidding strategy was presented for aggregated prosumers in [19] through the implementation of demand response. Further, a robust optimization-based day-ahead optimal bidding and scheduling method considering the dual uncertainty of renewable generations and the responsiveness of the demand side was proposed in [20] to maximize the profit of the aggregator. From the perspective of the overall benefit, a game-theoretic multi-follower bi-level programming was presented in [21] for the economic operation of the distribution network, which minimized the overall operation cost of the distribution network and maximized the profit of aggregated prosumers from exchanging power. Finally, a Stackelberg game model was proposed in [22] to peruse the competition among multiple aggregators and the distribution system operator. The equilibrium of the game model showed that each aggregator was able to maximize its profit in the market.

In summary, the positive effect of the aggregator on the benefit of aggregated prosumers and the microgrid is well studied and verified in the existing research. However, as more prosumers are involved, the massive amount of flexible energy resources of aggregated prosumers will endow the aggregator with comparatively strong market power [23], which was further exploited to perform price anticipation for excess profit [24]. As a result, the market power exploitation behavior of the aggregator will skew the market price away from the social optimum, harming the interests of non-aggregated prosumers, undermining the fairness of energy trading, resulting in market failure, market inefficiency, or market instability [25]. For example, results in [26] showed that when price anticipation was implemented in the trading strategy of the aggregator, the overall benefit of the community

microgrid might suffer losses. To further understand the formation and the negative effect of the market power, a bi-level optimization problem was formulated in [27] to quantify the excess profit obtained by the aggregator through strategic curtailment of generation in the electricity market. Similarly, a robust scheduling and bidding strategy for the price-making aggregator was presented in [28], showing that the aggregator influenced the market price and obtained higher profit by controlling its offered capacities. Then, a Nash Bargaining model was presented in [29] to describe the long-term single-leader multi-follower relationship between the aggregator and storage units, showing that the price anticipatory behavior of the aggregator might result in a socially suboptimal overall benefit. Finally, the strategic behavior of price-making aggregators was investigated in [30], showing that the maximized profit of the aggregator may not conclude the maximization of social welfare.

In general, previous research primarily concentrated on the Stackelberg competition market structure, which is a specific type of non-cooperative game that applies to P2P energy trading scenarios characterized by sequential decision-making and complete information (i.e., the aggregator formulates the trading price based on a comprehensive understanding of the cost and profit structures of prosumers, and subsequently each aggregated prosumer determines their energy trading quantity in response to the aggregator's price). However, in the more generalized P2P energy trading scenarios, where both the aggregator and individual prosumers (i.e., prosumers that are not engaged in aggregation) are involved in the market, each market participant will make their own decisions simultaneously due to the absence of a hierarchical order or dominant position. Meanwhile, predicting the price response characteristics of individual prosumers will also be challenging for the aggregator. Therefore, the existing Stackelberg competition model, which primarily focuses on leader-follower relationships, is not applicable for analyzing the aggregator's market power in the generalized P2P market. Instead, it is essential to develop a comprehensive non-cooperative competition model that encompasses both the aggregator and individual prosumers under the assumption of independent competition and simultaneous market clearing [31]. Furthermore, detailed research should also be conducted on the formation and the negative effect of the market power arising from prosumer aggregation. To fill the above research gaps, a game-theoretic method for market power modeling and restraint of aggregated prosumers is proposed in this paper. The main contributions are as follows: 1) A framework for the P2P energy trading market involving the aggregator and individual prosumers is established based on the simultaneous market clearing mechanism; 2) The competition model between the aggregator and individual prosumers is formulated as a non-cooperative game; 3) The impacts of market power on market price, trading profit, and overall benefit are further analyzed based on the non-cooperative game model;

4) The tentative offers penalty is introduced in the market to prohibit market power perception by suppressing market price fluctuations, and further restrain market power abuse in energy trading.

2. Peer-to-peer energy trading market with aggregator

The structure of the community microgrid studied in this paper is shown in Fig. 1. The energy coordination among prosumers can be achieved with the P2P energy trading market. Driven by self-interest, each prosumer will evaluate their energy supply or demand capability, then carry out energy trading accordingly. The aggregator will act as a market broker between small-scale prosumers and the P2P market, performing unified energy management and market bidding.

A P2P market operator (P2PMO) is introduced in the community microgrid to better coordinate energy trading among prosumers. During each trading period, the P2PMO will generate and publish the market clearing price (MCP) according to the bid information submitted by each prosumer. Each prosumer will adjust their energy production and consumption behavior according to the MCP for optimal energy trading.

2.1. Market model

Let $\mathcal{N} = \{1, 2, \dots, N\}$ denotes the set of prosumers in the decentralized market, indexed by $i \in \mathcal{N}$, and $N \triangleq |\mathcal{N}|$ is the total number of prosumers in the decentralized market. Let $\mathcal{T} = \{1, 2, \dots, T\}$ denotes the set of all trading periods of each day with $t \in \mathcal{T}$, and $T \triangleq |\mathcal{T}|$ is the total number of trading periods. In this study, it is considered that $T = 96$ and the time interval of each period is $\Delta t = 15 \text{ min}$.

2.1.1. Market clearing model

The uniform pricing mechanism is adopted in the decentralized market, in which each prosumer iteratively submits its trading function Eq.(1) to P2PMO during trading periods for market clearing [32]. Specifically, the energy price elasticity (EPE) coefficient $a_i^{(n)} > 0$ and energy purchasing intention (EPI) coefficient $b_i^{(n)}$ will be transferred to the P2PMO in the n -th iteration. It is worth mentioning that $a_i^{(n)}$ is an inherent characteristic of each prosumer i , which represents the price sensitivity and the energy dispatching flexibility. Therefore, EPE will be a constant $a_i^{(n)} = a_i$ during the iterative bidding of a specific trading period.

$$P_i = a_i c_{\text{MCP}}^{(n)} + b_i^{(n)} \quad (1)$$

where, P_i is the energy trading quantity of the prosumer i , the positive value of which denotes energy selling, while the negative value denotes energy purchasing; $c_{\text{MCP}}^{(n)}$ is the MCP in the n -th iteration.

The market clearing condition is that the total supply meets the total demand, as shown in Eq.(2). In the n -th iteration, after receiving the a_i and $b_i^{(n)}$ from each prosumer, P2PMO will generate and publish the MCP according to Eq.(3). Each prosumer will take the received MCP as a basis of pricing in the next iteration.

$$\sum_{i \in \mathcal{N}} P_i = 0 \Rightarrow \sum_{i \in \mathcal{N}} (a_i c_{\text{MCP}}^{(n)} + b_i^{(n)}) = 0 \quad (2)$$

$$c_{\text{MCP}}^{(n)} = - \sum_{i \in \mathcal{N}} b_i^{(n)} / \sum_{i \in \mathcal{N}} a_i \quad (3)$$

When the market reaches equilibrium, the MCP will converge to $c_{\text{MCP}}^{(n)} \rightarrow c_{\text{MCP}}^*$. The energy trading quantity $P_i \rightarrow P_i^*$ of each prosumer will be calculated with Eq.(1). If the $P_i^* > 0$, the prosumer i will sell energy P_i^* to

the market with a trading revenue of $c_{\text{MCP}}^* P_i^*$; Otherwise, it will buy energy $-P_i^*$ from the market with an expense of $-c_{\text{MCP}}^* P_i^*$. If the prosumer i fails to produce or consume the specified amount of energy as agreed, the deficiency part will be balanced by the utility grid, and the additional energy balancing cost will be punished on the default prosumer.

2.1.2. Market power model

Market power refers to the ability of prosumers to drive the market price over a competitive level [33]. According to Eq.(3), the MCP is determined by the a_i and $b_i^{(n)}$ of each prosumer i . Therefore, we define $\mu_i \geq 0$ as the price impact factor of the prosumer i , which represents the rate of MCP deviation Δc_{MCP} with respect to the trading strategy adjustment Δb_i of the prosumer i as follows:

$$\Delta c_{\text{MCP}} = - \mu_i \frac{\Delta b_i}{a_i} \quad (4)$$

As a result, when the prosumer i performs price anticipation during the iterative bidding, the anticipated MCP of the prosumer i is shown as follows:

$$\hat{c}_{\text{MCP},i}^{(n)} = c_{\text{MCP}}^{(n-1)} - \frac{\mu_i}{a_i} (b_i^{(n)} - b_i^{(n-1)}) \quad (5)$$

2.2. Prosumer model

In iterative bidding, each prosumer will determine the optimal trading strategy to maximize its profit. To be specific, the EPI coefficient $b_i^{(n)}$ will be optimized as follows:

$$\max_{b_i^{(n)}} \pi_i = M_i - C_i \quad (6)$$

where, M_i is the trading revenue of the prosumer i ; C_i is the trading cost of the prosumer i , in which both energy resource dispatching cost and opportunity cost (i.e., additional costs the prosumer i will have to pay in the future due to the energy trading activities conducted in the current time period) are considered.

The market power of individual prosumers is relatively weak due to trivial market share and limited market perception ability. Therefore, individual prosumers will regard themselves as price takers in the market, i.e., the MCP will not be affected by their trading strategies, and thus the price impact factor is regarded as $\mu_i = 0$. Then, we have the trading revenue of prosumer i as follows:

$$P_i = a_i c_{\text{MCP}}^{(n-1)} + b_i^{(n)} \quad (7)$$

$$M_i = \hat{c}_{\text{MCP}}^{(n)} P_i \approx c_{\text{MCP}}^{(n-1)} P_i \quad (8)$$

The trading cost of individual prosumers is defined as a quadratic function Eq.(9), and the marginal trading cost of individual prosumers is derived as a linear function Eq.(10). It is worth noting that both variable costs (e.g. controllable DERs generation cost) and fixed costs (e.g. battery storage charging and discharging cost) are considered in the cost function.

$$C_i = c_i P_i^2 + d_i P_i \quad (9)$$

$$MC_i = 2c_i P_i + d_i \quad (10)$$

where, $c_i > 0$ and $d_i > 0$ are the variable cost (VC) and fixed cost (FC) coefficients, respectively.

Finally, we have the objective function of individual prosumer pricing as follows:

$$\pi_i = M_i - C_i = c_{\text{MCP}}^{(n-1)} P_i - (c_i P_i^2 + d_i P_i) \quad (11)$$

The constraint on the EPI coefficient $b_i^{(n)}$ is considered in the prosumer model.

$$\Rightarrow \left(P_i^{\min} - a_i c_{\text{MCP}}^{(n-1)} \right) \leq b_i^{(n)} \leq \left(P_i^{\max} - a_i c_{\text{MCP}}^{(n-1)} \right) \quad (12)$$

where $P_i^{\max}$ and $P_i^{\min}$ are the upper and lower bound of P_i , which is determined by the energy production and consumption capability of each energy resource of the prosumer i .

2.3. Aggregator model

Let $\mathcal{L}_g = \{1, 2, \dots, L_g\} \subseteq \mathcal{N}$ denotes the set of prosumers participating in the broker service provided by the aggregator g , and $L_g \triangleq |\mathcal{L}_g|$ denotes the total number of prosumers in $\mathcal{L}_g$. The aggregator g will uniformly manage the energy resources of each prosumer $j \in \mathcal{L}_g$. The trading cost of the aggregator can be obtained by super-positioning the marginal cost function MC_j of each individual prosumer $j \in \mathcal{L}_g$ as follows:

$$\begin{aligned} MC_j &= 2c_j P_j + d_j \\ \Rightarrow P_g &= \sum_{j \in \mathcal{L}_g} P_j = MC_g \sum_{j \in \mathcal{L}_g} \frac{1}{2c_j} - \sum_{j \in \mathcal{L}_g} \frac{d_j}{2c_j} \end{aligned} \quad (13)$$

$$\begin{aligned} \Rightarrow MC_g &= 2 \left(\frac{1}{\sum_{j \in \mathcal{L}_g} \frac{1}{c_j}} \right) P_g + \left(\sum_{j \in \mathcal{L}_g} \frac{d_j}{c_j} / \sum_{j \in \mathcal{L}_g} \frac{1}{c_j} \right) \\ C_g &= \left(\frac{1}{\sum_{j \in \mathcal{L}_g} \frac{1}{c_j}} \right) P_g^2 + \left(\sum_{j \in \mathcal{L}_g} \frac{d_j}{c_j} / \sum_{j \in \mathcal{L}_g} \frac{1}{c_j} \right) P_g + \delta_g \end{aligned} \quad (14)$$

where, C_g is the trading cost of the aggregator g ; P_g is the energy trading quantity of the aggregator g , the positive value of which denotes energy selling, while the negative value denotes energy purchasing; c_g and d_g are the VC and FC coefficients of the aggregator, respectively; $\delta_g \leq 0$ indicates that the aggregator can reduce the costs of prosumers by unified energy management even when it does not participate in energy trading (i.e., $P_g = 0$).

The market power of the aggregator g is significant compared with individual prosumers $i \notin \mathcal{L}_g$. Therefore, the price impact factor $\mu_g > 0$ is considered in the trading price $\hat{c}_{\text{MCP},g}^{(n)}$. Then, the trading revenue of aggregator g will be determined as follows:

$$P_g = a_g c_{\text{MCP}}^{(n-1)} + b_g^{(n)} \quad (15)$$

$$M_g = \hat{c}_{\text{MCP},g}^{(n)} P_g = \left(c_{\text{MCP}}^{(n-1)} - \frac{\mu_g}{a_g} (b_g^{(n)} - b_g^{(n-1)}) \right) P_g \quad (16)$$

where, M_g is the trading revenue of the aggregator g ; a_g and $b_g^{(n)}$ are the EPE and EPI of the aggregator g in the n -th iteration.

The aggregator will participate in the market trading as a virtual prosumer. Therefore, the trading strategy of the aggregator is similar to that of individual prosumers as follows:

$$\begin{aligned} \max_{b_g^{(n)}} \pi_g &= M_g - C_g \\ &= \left(c_{\text{MCP}}^{(n-1)} - \frac{\mu_g}{a_g} (b_g^{(n)} - b_g^{(n-1)}) \right) P_g - (c_g P_g^2 + d_g P_g) \end{aligned} \quad (17)$$

The constraints on the EPI coefficient $b_g^{(n)}$ of the aggregator g is as follows:

$$\left(P_g^{\min} - a_g c_{\text{MCP}}^{(n-1)} \right) \leq b_g^{(n)} \leq \left(P_g^{\max} - a_g c_{\text{MCP}}^{(n-1)} \right) \quad (18)$$

where, $P_g^{\max}$ and $P_g^{\min}$ are the upper and lower bound of P_g , which are determined by the energy production and consumption capability of each prosumer $j \in \mathcal{L}_g$.

3. Game-theoretic approach to market power modeling and restraint

In this section, the competition between the aggregator g and individual prosumers $i \notin \mathcal{L}_g$ will be modeled as a non-cooperative game. Then, the impact of market power on price formation, trading profit, and overall benefit will be analyzed. Finally, a method to restrain market power will be proposed to relieve the impact of market power on the trading fairness and efficiency of the market.

3.1. Non-cooperative game model

In this subsection, both the aggregator g and individual prosumers $i \notin \mathcal{L}_g$ are generalized to prosumers i with market power, and Eq.(5) is adopted as the price anticipating method in the trading strategy of each prosumer. Specifically, if the prosumer i represents the aggregator, then we have $\mu_i > 0$; otherwise, we have $\mu_i = 0$.

3.1.1. Non-cooperative game formulation

For the non-cooperative game among prosumers in a P2P energy market, we have:

Player: All prosumers in the community microgrid.

Strategy: Each prosumer controls their EPI coefficient $b_i^{(n)}$.

Payoff: The utility of prosumer i is given in Eq.(19). $\mathbf{b}_{-i}^*$ denotes the set of strategies except for prosumer i .

$$\begin{aligned} \pi_i &= \left(b_i^{(n)} \mathbf{b}_{-i}^{(n)} \right) \\ &= \left(c_{\text{MCP}}^{(n-1)} - \frac{\mu_i}{a_i} (b_i^{(n)} - b_i^{(n-1)}) \right) P_i - (c_i P_i^2 + d_i P_i) \end{aligned} \quad (19)$$

The set of all possible EPIs of the prosumer $i \in \mathcal{N}$ is denoted by Eq. (20).

$$S_i = \left\{ b_i^{(n)} \mid \left(P_i^{\min} - a_i c_{\text{MCP}}^{(n-1)} \right) \leq b_i^{(n)} \leq \left(P_i^{\max} - a_i c_{\text{MCP}}^{(n-1)} \right) \right\} \quad (20)$$

For each prosumer $i \in \mathcal{N}$, the optimal trading strategy of EPI is

$$b_i^{(n)} = \underset{b_i^{(n)}}{\operatorname{argmax}} \pi_i \left(b_i^{(n)}, \mathbf{b}_{-i}^{(n)} \right) \quad (21)$$

For the P2PMO, according to the a_i and $b_i^{(n)}$ from each prosumer, the updated MCP is

$$c_{\text{MCP}}^{(n)} = - \sum_{i \in \mathcal{N}} b_i^{(n)} / \sum_{i \in \mathcal{N}} a_i \quad (22)$$

It is worth noting that for the aggregator with market power, we have $\mu_i > 0$ in Eq.(19) and Eq.(21). For price-taking individual prosumers, we have $\mu_i = 0$. Then, the bidding process of the non-cooperative game in the P2P energy market is described in Algorithm 1.

Algorithm 1. The bidding process of the non-cooperative game in the P2P energy market

Input: a_i, c_i and d_i of each prosumer

Output: Equilibrium MCP c_{MCP}^* of the P2P energy market and b_i^* of each prosumer i

Assign the utility grid retail price c_{retail} as the initial MCP $c_{MCP}^{(0)}$; $-a_i c_{\text{retail}}$ as the initial EPI $b_i^{(0)}$; $n = 0$;

do

$n = n + 1$;

for all prosumer $i \in \mathcal{N}$ **do**

Compute $b_i^{(n)}$ with $c_{MCP}^{(n-1)}$ and $b_i^{(n-1)}$ according to Eq.(21);

end for

Compute the updated MCP $c_{MCP}^{(n)}$ according to Eq.(22);

while $(|c_{MCP}^{(n)} - c_{MCP}^{(n-1)}| > \varepsilon_{MCP})$

$c_{MCP}^* = c_{MCP}^{(n)}$; $b_i^* = b_i^{(n)}$

where, $\varepsilon_{MCP} = 0.001$ is the convergence coefficient.

The convergence of the above algorithm depends on the EPE coefficient a_i of each prosumer. When a_i is too small, minor fluctuations in market price $c_{MCP}^{(n)}$ will result in a dramatic change in the decision value of $b_i^{(n)}$, which may lead to oscillation or divergence of the algorithm [25]. However, when a_i meets certain conditions, we can prove the convergence of the algorithm. Here we investigate a particular condition $a_i \geq 1/2c_i$, which gives clear economic meaning to the trading function Eq.(1) of each prosumer (i.e., the trading function has higher or equal elasticity than the marginal cost function).

Proof. We use Banach's fixed point theorem to prove the convergence of Algorithm 1. According to the theorem, let X be a complete metric space (X, d) and $f : X \rightarrow X$ be a contraction mapping. Then, f has a unique fixed point x^* .

In our case, the metric space X is $\mathcal{R}$ with the associated metric being the absolute value of the difference $d(x_1, x_2) = |x_1 - x_2|$. Then, f is a contraction mapping with parameter λ on X if $\exists 0 \leq \lambda < 1$ such that:

$$\forall x_1, x_2 \in X : d(f(x_1), f(x_2)) \leq \lambda d(x_1, x_2) \quad (23)$$

In Algorithm 1, we have

$$|f(x_1) - f(x_2)| = \lambda |x_1 - x_2| \quad (24)$$

$$\lambda = \left| \sum_{i \in \mathcal{N}'} a_i \left(1 - \frac{1}{(2a_i c_i + \mu_i)} \right) / \sum_{i \in \mathcal{N}'} a_i \right| \quad (25)$$

The above formula shows that given $c_i > 0$, $\mu_i \geq 0$ and condition $a_i \geq 1/2c_i$, we have $0 \leq \lambda < 1$ in Eq.(24) and thus f is a contraction mapping. By Banach's fixed point theorem, the recursion in Algorithm 1 converges to a unique fixed point x^* .

3.1.2. Non-cooperative game equilibrium

We define that the Nash equilibrium of the non-cooperative game is a set of EPI strategies of all prosumers such that no prosumer deviates unilaterally.

Definition 1 (Nash equilibrium). The Nash equilibrium is a set of EPI for all prosumers such that

$$\pi_i(b_i^*, \mathbf{b}_{-i}^*) \geq \pi_i(b_i, \mathbf{b}_{-i}^*), \forall i \in \mathcal{N}, \forall b_i \in S_i \quad (26)$$

Proposition 1. A unique Nash equilibrium exists in the non-cooperative game of the market.

Proof. $\pi_i(b_i, \mathbf{b}_{-i}^*)$ is a continuous function in b_i . Taking the second derivative of the prosumer utility function $\pi_i(b_i, \mathbf{b}_{-i}^*)$ with respect of b_i ,

we have

$$\frac{\partial^2 \pi_i}{\partial b_i^2} = -2(c_i + \mu_i/a_i) < 0 \quad (27)$$

Eq. (27) show that for $\pi_i(b_i, \mathbf{b}_{-i}^*)$ the second derivative is strictly negative due to $a_i > 0$, $c_i > 0$ and $\mu_i \geq 0$ for $\forall i \in \mathcal{N}$. Thus, $\pi_i(b_i, \mathbf{b}_{-i}^*)$ is a strictly concave function.

Given the strategy space defined by Eq.(20), S_i is a non-empty, convex, and compact subset of Euclidean space $\mathcal{R}$. From the above, we can conclude that a non-cooperative game among prosumers is equivalent to a strictly concave N -player game. Thus, a unique Nash equilibrium exists among prosumers in the market.

3.2. Non-cooperative game properties

3.2.1. Impact of market power on price formation

1) Market price formation of a perfectly competitive market.

For comparison, we first investigate the case under a perfectly competitive market, where each prosumer independently participates in energy trading. All prosumers are price takers in this market due to the trivial market power and enormous market scale. The price impact factor of each prosumer is assumed as $\mu_i = 0$. According to Eq.(21), when $c_{MCP}^{(n)} \rightarrow c_{MCP}^*$ and $b_i^{(n)} \rightarrow b_i^*$, we have:

$$b_i^* = \underset{b_i}{\operatorname{argmax}} \pi_i(b_i, \mathbf{b}_{-i}) = \frac{(1 - 2a_i c_i) c_{MCP}^* - d_i}{2c_i} \quad (28)$$

Solve the trading strategy of each prosumer uniformly, i.e., substituting Eq.(28) into the market clearing condition Eq.(3), we have the equilibrium MCP as follows:

$$\sum_{i \in \mathcal{N}'} b_i^* = \sum_{i \in \mathcal{N}'} \frac{1 - 2a_i c_i}{2c_i} c_{MCP}^* - \sum_{i \in \mathcal{N}'} \frac{d_i}{2c_i} \quad (29)$$

$$c_{MCP}^* = - \sum_{i \in \mathcal{N}'} b_i^* / \sum_{i \in \mathcal{N}'} a_i \Rightarrow c_{MCP}^* = \sum_{i \in \mathcal{N}'} \frac{d_i}{c_i} / \sum_{i \in \mathcal{N}'} \frac{1}{c_i} \quad (30)$$

The formula above shows that the MCP of the perfect competition market is determined by energy resources dispatching cost coefficients (c_i and d_i) and irrelevant to trading preference coefficients (a_i and b_i) of each prosumer. As a comparison, when the community microgrid adopts unified energy management (i.e., each prosumer submits their genuine c_i and d_i to the P2PMO for market clearing, and the P2PMO will optimize the energy exchange between prosumers for the lowest dispatching cost), the MCP will be as follows:

$$\begin{aligned}
c_{\text{MCP}}^* &= 2c_i P_i + d_i \\
\Rightarrow \sum_{i \in \mathcal{I}'} P_i &= c_{\text{MCP}}^* \sum_{i \in \mathcal{I}'} \frac{1}{2c_i} - \sum_{i \in \mathcal{I}'} \frac{d_i}{2c_i} = 0 \\
\Rightarrow c_{\text{MCP}}^* &= \frac{\sum_{i \in \mathcal{I}'} d_i}{\sum_{i \in \mathcal{I}'} c_i}
\end{aligned} \quad (31)$$

The formula above shows that the MCP of the perfect competition market is consistent with that of unified energy management. This conclusion is consistent with the equilibrium characteristics of a perfect competition market, i.e., each profit-maximizing prosumer faces a market price determined by its marginal cost or marginal utility, and the market is allocative efficient [31].

2) Market price formation of a monopoly market.

Then, we consider the case under a monopoly market, where prosumers $j \in \mathcal{L}_g$ participate in the broker service of the aggregator g , while other prosumers $i \notin \mathcal{L}_g$ independently participate in energy trading. Compared with individual prosumers, the aggregator is a market entity with significant market power. Therefore, the price impact factor of the aggregator g will be considered in the optimal pricing strategy as follows:

$$\begin{aligned}
b_g^* &= \frac{(1 - 2a_g(c_g + \mu_g/2a_g))c_{\text{MCP}}^* - d_g}{2(c_g + \mu_g/a_g)} + \frac{b_g^*(\mu_g/a_g)}{2(c_g + \mu_g/a_g)} \\
\Rightarrow b_g^* &= \frac{(1 - 2a_g(c_g + \mu_g/2a_g))c_{\text{MCP}}^* - d_g}{2(c_g + \mu_g/2a_g)}
\end{aligned} \quad (32)$$

Similar to Eq.(29)–(30), we have the equilibrium MCP as follows:

$$c_{\text{MCP}}^* = \frac{\frac{d_g}{c_g + \mu_g/2a_g} + \sum_{i \notin \mathcal{L}_g} \frac{d_i}{c_i}}{\frac{1}{c_g + \mu_g/2a_g} + \sum_{i \notin \mathcal{L}_g} \frac{1}{c_i}} \quad (33)$$

We can draw the following conclusions from the above formula:

Remark 1. The MCP is affected by the price anticipation of the aggregator instead of the prosumer aggregation.

Eq.(33) shows that the substantial reason for the MCP to deviate from the MCP of the perfect competition market is that the aggregator considered the price impact factor μ_g in the trading strategy of the aggregator g . On the contrary, when the market power is neglected (i.e., the price impact factor of the aggregator is taken as $\mu_g = 0$), the MCP will remain unchanged as follows, even though the prosumer aggregation still exists.

$$\begin{aligned}
c_{\text{MCP}}^* &= \left(\frac{d_g}{c_g} + \sum_{i \in \mathcal{L}_g} \frac{d_i}{c_i} \right) / \left(\frac{1}{c_g} + \sum_{i \in \mathcal{L}_g} \frac{1}{c_i} \right) = \left(\sum_{j \in \mathcal{L}_g} \frac{d_j}{c_j} + \sum_{i \notin \mathcal{L}_g} \frac{d_i}{c_i} \right) / \left(\sum_{j \in \mathcal{L}_g} \frac{1}{c_j} + \sum_{i \notin \mathcal{L}_g} \frac{1}{c_i} \right) \\
&= \sum_{i \in \mathcal{I}'} \frac{d_i}{c_i} / \sum_{i \in \mathcal{I}'} \frac{1}{c_i}
\end{aligned} \quad (34)$$

Remark 2. The aggregator can manipulate the market price by adjusting the trading quantity.

When the market reaches equilibrium, the energy trading quantity P_g^* of the aggregator g is deduced from Eq.(1) as follows:

$$\begin{aligned}
P_g^* &= a_g c_{\text{MCP}}^* + b_g^* \\
&= \frac{1}{2(c_g + \mu_g/a_g)} \frac{\sum_{k \notin \mathcal{L}_g} \frac{d_k - d_g}{c_k}}{\frac{1}{c_g + \mu_g/a_g} + \sum_{k \notin \mathcal{L}_g} \frac{1}{c_k}}
\end{aligned} \quad (35)$$

For comparison, take the first derivative of the MCP with respect to μ_g , we have:

$$\frac{\partial c_{\text{MCP}}^*}{\partial \mu_g} = P_g^* \frac{\frac{a_g(c_g + \mu_g/a_g)}{c_g + \mu_g/a_g}}{\frac{1}{c_g + \mu_g/a_g} + \sum_{k \notin \mathcal{L}_g} \frac{1}{c_k}} \quad (36)$$

Comparing Eq.(35) and Eq.(36), it is evident that when the aggregator sells energy to the market $P_g^* > 0$, it can manipulate the MCP to rise for higher profit by performing energy trading quantity curtailment and vice versa. This conclusion is consistent with the equilibrium characteristics of a monopoly market, i.e., the aggregator sells lower quantities of energy at a higher price [27]. At this point, the profit of individual prosumers and the overall benefit of the microgrid may be negatively affected. The adverse consequences will be further analyzed in the following sections.

3.2.2. Impact of market power on trading profit

1) Impact of market power on the profit of the aggregator.

We consider the case under the monopoly market in section 3.2.1. The trading profit π_g^* of the aggregator g is deduced from Eq.(17) as follows.

$$\pi_g^* = c_{\text{MCP}}^* P_g^* - (c_g P_g^{*2} + d_g P_g^*) = P_g^* (c_{\text{MCP}}^* - d_g) - c_g P_g^{*2} = (c_g + \mu_g/a_g) P_g^{*2} \quad (37)$$

Then, we have the following proposition:

Proposition 2. Let μ_g be the price impact factor of the aggregator g , then we have:

$$\mu_g = a_g / \sum_{i \notin \mathcal{L}_g} \frac{1}{2c_i} \quad (38)$$

Proof. According to the market clearing condition Eq.(2) and trading strategy Eq.(15), we have:

$$\begin{aligned}
P_g &= - \sum_{i \notin \mathcal{L}_g} P_i \\
\Rightarrow a_g c_{\text{MCP}}^{(k-1)} + b_g^{(k)} &= - \sum_{i \notin \mathcal{L}_g} (a_i c_{\text{MCP}}^{(n)} + b_i^{(n)}) \\
&= - \sum_{i \notin \mathcal{L}_g} \left(a_i \left(c_{\text{MCP}}^{(n-1)} - \frac{\mu_g}{a_g} (b_g^{(n)} - b_g^{(n-1)}) \right) + b_i^{(n)} \right)
\end{aligned} \quad (39)$$

Take the first derivative of Eq.(39) on both sides with respect of d_g , we have:

$$\begin{aligned}
\frac{\partial b_g^{(n)}}{\partial d_g} &= - \sum_{i \notin \mathcal{L}_g} \frac{1}{2c_i} \left(- \frac{\mu_g}{a_g} \right) \frac{\partial b_g^{(n)}}{\partial d_g} \\
\Rightarrow \mu_g &= a_g / \sum_{i \notin \mathcal{L}_g} \frac{1}{2c_i}
\end{aligned} \quad (40)$$

Proposition 3. Let $\hat{\mu}_g$ be the estimated price impact factor of the aggregator g , and $\pi_g^*(b_g^*, \hat{\mu}_g)$ be the profit of the aggregator g at the equilibrium of the non-cooperative game. Then, we have:

$$\pi_g^*(b_g^*, \mu_g) \geq \pi_g^*(b_g^*, \hat{\mu}_g), \forall \hat{\mu}_g > 0 \quad (41)$$

Proof. $\pi_g^*(b_g^*, \hat{\mu}_g)$ is a continuous function in $\hat{\mu}_g$. Take the first derivative of the equilibrium trading profit function $\pi_g^*(b_g^*, \hat{\mu}_g)$ with respect of $\hat{\mu}_g$ as below.

$$\frac{\partial \pi_g^*}{\partial \mu_g} = \frac{1}{4a_g} \frac{\left(\sum_{k \notin \mathcal{L}_g} \frac{d_k - d_g}{c_k} \right)^2}{\left(\sum_{k \notin \mathcal{L}_g} \frac{c_g + \mu_g / 2a_g}{c_k} + 1 \right)^3} (1 - \hat{\mu}_g / \mu_g) \quad (42)$$

The above formula shows that given positive a_g , c_g , μ_g and c_k , π_g^* will monotonically increase when $\hat{\mu}_g < \mu_g$ due to the constant positive first derivative, and reach maximum when $\hat{\mu}_g = \mu_g$. Then, π_g^* will monotonically decrease to negative when $\hat{\mu}_g > \mu_g$.

Finally, we can draw the following conclusions from the above proposition:

Remark 3. Individual prosumers are inherently motivated to perform prosumer aggregation.

In Remark 1, we concluded that prosumer aggregation will not affect the MCP if market power is neglected in the trading strategy of the aggregator. At this point, the total trading revenue of prosumers $j \in \mathcal{L}_g$ will be equal to that of the aggregator g when selling the same quantity of energy to the market. Meanwhile, the trading cost of the aggregator g is lower than or equal to the total trading cost of individual prosumers $j \in \mathcal{L}_g$ due to unified management, indicating that the aggregator will obtain a higher profit. Therefore, we can draw the following conclusion from Proposition 3, showing that individual prosumers $j \in \mathcal{L}_g$ are inherently motivated to perform prosumer aggregation due to profitability.

$$\pi_g^*(b_g^*, \mu_g^*) \geq \pi_g^*(b_g^*, 0) \geq \sum_{j \in \mathcal{L}_g} \pi_j^*(b_j^*, 0) \quad (43)$$

Remark 4. The aggregator requires an accurate estimation of the price impact factor to maximize profit.

Proposition 3 shows that the aggregator g will obtain the maximized profit only when the price impact factor is accurately estimated as $\hat{\mu}_g = \mu_g$. In a particular case, when the price impact factor is underestimated, the aggregator may not get the maximized profit. Conversely, when the price impact factor is overestimated, the aggregator may even sustain losses.

Remark 5. The a_g of the aggregator and the a_i of each prosumer $i \notin \mathcal{L}_g$ does not affect the market equilibrium.

From the perspective of market equilibrium, it is evident in Eq.(33) that the a_i of prosumers $i \notin \mathcal{L}_g$ does not affect the MCP. Then, we investigate the impact of a_g on the MCP.

We can conclude from Eq.(38) that the estimated price impact factor $\hat{\mu}_g$ is proportional to a_g as in Eq.(44). Then, substituting $\hat{\mu}_g$ into Eq.(33), we have:

$$\hat{\mu}_g = a_g / \sum_{i \notin \mathcal{L}_g} \frac{1}{2\hat{c}_i} \quad (44)$$

$$\begin{aligned} \hat{c}_{-g} &= \frac{\hat{\mu}_g}{2a_g} = 1 / \sum_{i \notin \mathcal{L}_g} \frac{1}{\hat{c}_i} \\ \Rightarrow c_{\text{MCP}}^* &= \frac{\frac{d_g}{c_g + \hat{c}_{-g}} + \sum_{i \notin \mathcal{L}_g} \frac{d_i}{c_i}}{\frac{1}{c_g + \hat{c}_{-g}} + \sum_{i \notin \mathcal{L}_g} \frac{1}{c_i}} \end{aligned} \quad (45)$$

where, $\hat{c}_i$ is the estimated VC of the prosumer $i \notin \mathcal{L}_g$; $\hat{c}_{-g}$ is the estimated elasticity of the market.

The above formula shows that c_{MCP}^* is irrelevant to a_g when the

market reaches equilibrium. Similarly, we can also assert that the trading power and trading profit of each prosumer will not be affected by a_g and a_i .

2) Impact of market power on individual prosumers.

According to Eq.(11), the trading profit of individual prosumers $i \notin \mathcal{L}_g$ are as follows:

$$\pi_i^* = \frac{1}{4c_i} \frac{\left(\frac{d_g - d_i}{c_g + \mu_g / 2a_g} + \sum_{k \notin \mathcal{L}_g} \frac{d_k - d_i}{c_k} \right)^2}{\left(\frac{1}{c_g + \mu_g / 2a_g} + \sum_{k \neq i} \frac{1}{c_k} \right)^2} \quad (46)$$

$$\begin{aligned} \frac{\partial \pi_i^*}{\partial \mu_g} &= (c_{\text{MCP}}^* - d_i) \left(\frac{1}{c_g + \mu_g / 2a_g} + \sum_{k \notin \mathcal{L}_g} \frac{1}{c_k} \right) \left(\sum_{k \notin \mathcal{L}_g} \frac{d_k - d_g}{c_k} \right) \\ &= P_i^* P_g^* \left(4c_i (c_g + \mu_g / 2a_g) \left(\frac{1}{c_g + \mu_g / 2a_g} + \sum_{k \notin \mathcal{L}_g} \frac{1}{c_k} \right)^2 \right) \end{aligned} \quad (47)$$

The formula above shows that the individual prosumer $i \notin \mathcal{L}_g$ may either benefit or suffer loss from the involvement of the aggregator g , which depends on their fixed cost coefficients d_i . Specifically, when the aggregator g sells (purchases) energy $P_g > 0$ to the market, the individual prosumer $i \notin \mathcal{L}_g$ will benefit from the market power of the aggregator when it also sells (purchases) energy to the market, i.e., $P_i > 0$ when the fixed cost coefficient d_i is lower (higher) than c_{MCP}^* , which is commonly known as the 'hitchhiking effect'. On the contrary, energy purchasing (selling) prosumers will suffer losses. Therefore, we have the following conclusion:

Remark 6. The market power of the aggregator g may result in unfair distribution of welfare among prosumers.

3.2.3. Impact of market power on overall benefit of microgrid

1) Impact of market power on overall trading quantity

Definition 2 (Overall trading quantity). The overall trading quantity between the aggregator g and individual prosumers $i \notin \mathcal{L}_g$ at the equilibrium of the non-cooperative game, denoted by $Q(\mathbf{b}^*, \mu_g)$, is defined as

$$Q(\mathbf{b}^*, \mu_g) = |P_g| \quad (48)$$

Proposition 4. Let μ_g be the price impact factor of the aggregator g . Then, we have

$$Q(\mathbf{b}^*, \mu_g) \langle Q(\mathbf{b}^*, 0), \forall \mu_g > 0 \quad (49)$$

Proof. $Q(\mathbf{b}^*, \mu_g)$ is a continuous function in μ_g . Taking the first derivative of the square of trading profit function $Q^2(\mathbf{b}^*, \mu_g)$ with respect to μ_g as follows. Given positive a_g , c_g , μ_g and c_k for $\forall i \in \mathcal{N}$, we conclude that $Q^2(\mathbf{b}^*, \mu_g)$ monotonically decreases with respect to $\mu_g > 0$.

$$Q^2(\mathbf{b}^*, \mu_g) = \frac{1}{4(c_g + \mu_g / 2a_g)^2} \frac{\left(\sum_{k \notin \mathcal{L}_g} \frac{d_k - d_g}{c_k} \right)^2}{\left(\frac{1}{c_g + \mu_g / 2a_g} + \sum_{k \notin \mathcal{L}_g} \frac{1}{c_k} \right)^2} \quad (50)$$

Table 1

Typical configuration of energy resources for diversified prosumers in the community microgrid.

Parameters		Residential Prosumer (#1-#45)	Residential Prosumer (#46-#90)	Commercial Prosumer (#91-#96)	Commercial Prosumer (#97-#102)
Load	Power	2.8 kW	3 kW	20 kW	40 kW
	Operation Time Period	17:00–24:00	17:00–24:00	8:00–17:00	8:00–17:00
PV	Peak Power	2.5 kW	3 kW	8 kW	20 kW
	Capacity	7 kWh	13.5 kWh	55 kWh	150 kWh
ESS	Max. Charging & Discharging Power	3 kW	3.5 kW	12 kW	20 kW
	Levelized Cost Of Electricity (LCOE)	0.84 CNY/kWh	0.84 CNY/kWh	0.76 CNY/kWh	0.76 CNY/kWh

$$= -\frac{\frac{\partial Q^2(\mathbf{b}^* \mu_g)}{\partial \mu_g}}{4a_g(c_g + \mu_g/2a_g)^3} \left(\frac{\sum_{k \notin \mathcal{D}_g} \frac{1}{c_k} \right) \left(\frac{\sum_{k \notin \mathcal{D}_g} d_k - d_g}{c_k} \right)^2 < 0 \quad (51)$$

Proposition 4 shows that the overall trading quantity between the aggregator and individual prosumers will reduce when the aggregator takes advantage of its market power for profit, which is consistent with the conclusion in [Remark 2](#). The reduced energy trading quantity will weaken the energy resources allocation ability of the market, leading to a decreased level of peak load shifting and renewable energy self-consumption, which will bring challenges to the safe and reliable operation of the power grid.

2) Impact of market power on overall trading profit.

Definition 3 (Overall trading profit). The overall trading profit of aggregator g and individual prosumers $i \notin \mathcal{D}_g$ at the equilibrium of the non-cooperative game, denoted by $\Pi(\mathbf{b}^*, \mu_g)$, is defined as

$$\Pi(\mathbf{b}^*, \mu_g) = \pi_g^*(b_g^*, \mu_g) + \sum_{i \notin \mathcal{D}_g} \pi_i^*(b_i^*, 0) = (c_g + \mu_g/a_g)P_g^{*2} + \sum_{i \notin \mathcal{D}_g} c_i P_i^{*2} \quad (52)$$

Proposition 5. Let μ_g be the price impact factor of the aggregator g . Then, we have

$$\Pi(\mathbf{b}^*, \mu_g) \Pi(\mathbf{b}^*, 0), \forall \mu_g > 0 \quad (53)$$

Proof. $\Pi(\mathbf{b}^*, \mu_g)$ is a continuous function in μ_g . Take the first derivative of the trading profit function $\Pi(\mathbf{b}^*, \mu_g)$ with respect of μ_g as follows. Given positive a_g, c_g, μ_g and a_i, c_i, μ_i for $\forall i \in \mathcal{N}$, we conclude that $\Pi(\mathbf{b}^*, \mu_g)$ monotonically decreases with respect to $\mu_g > 0$.

$$= -\frac{\frac{\partial \Pi(\mathbf{b}^* \mu_g)}{\partial \mu_g}}{4a_g} \left(\frac{\sum_{k \notin \mathcal{D}_g} \frac{\mu_g/2a_g}{c_g} \right) \left(\frac{\sum_{k \notin \mathcal{D}_g} d_k - d_g}{c_k} \right)^2 < 0 \quad (54)$$

Proposition 5 shows that although the aggregator g can make an excess profit by performing price anticipation in energy trading, the overall economic benefit of the community microgrid will suffer losses. Moreover, since the uniqueness of Nash equilibrium is proved in [Proposition 1](#), the Price of Anarchy (PoA) of the market can be defined as the ratio between the equilibrium and the optimal microgrid overall economic benefit solution. Consequently, the PoA of the non-cooperative game will be $PoA > 1$ due to the lower overall economic benefit compared with the unified energy management. This conclusion is

Table 2

Retail price and feed-in tariff of the utility grid.

t	Corresponding Time Period	Price Period	Retail Price (CNY/kWh)	Feed-in Tariff (CNY/kWh)
[1,28], [93,96]	(0:00–7:00), (23:00–0:00)	Valley	0.443	0.370
[41,60], [73,84]	(10:00–15:00), (18:00–21:00)	Peak	1.095	0.370
[29,40], [61,72], [85,92]	(7:00–10:00), (15:00–18:00), (21:00–23:00)	Average	0.728	0.370

consistent with the equilibrium characteristics of a monopoly market, i. e., the monopoly market is less efficient than the perfect competition market [\[27\]](#).

3.3. Method to restrain market power

In [Section 3.2.1](#), we proved that prosumer aggregation itself would not affect the equilibrium of the market. However, when the aggregator g performs price anticipation for excess profit, the fair benefit distribution among prosumers as well as the overall benefit of the microgrid may be affected, as we analyzed in [Section 3.2.2](#) and [Section 3.2.3](#), respectively. In the worst case, prosumers $i \notin \mathcal{D}_g$ may also perform prosumer aggregation, forming a new aggregator h to counter the market power of the aggregator g , which will further harm the overall benefit of the microgrid as follows:

$$\Delta \Pi(\mathbf{b}^* \mu) = -\frac{(\mu_g/2a_g + \mu_h/2a_h)(d_g - d_h)^2}{(c_g + \mu_g/2a_g + c_h + \mu_h/2a_h)^3} \left(\frac{\Delta \mu_g}{4a_g} + \frac{\Delta \mu_h}{4a_h} \right) < 0 \quad (55)$$

where, $\mu = \{\mu_g, \mu_h\}$ is the set of price impact factors; μ_h is the price impact factor of the aggregator h .

Therefore, specific measures should be taken to prevent market power formation in energy trading. As introduced in [\[23\]](#), the aggregator g is able to perceive μ_g from historical trading records by observing the response of $c_{MCP}^{(n)}$ to the deliberately offered $b_i^{(n)}$ in the iterative bidding process. Therefore, a feasible method to prevent market power perception is to restrain market price fluctuation by prohibiting tentative offers during iterative bidding. In a real system, such penalty fees will be collected by the P2PMO to promote the fairness and efficiency of the market. In a particular case, a penalty as follows can be imposed on the prosumers when they submit tentative offers $b_i^{(n)} \neq b_i^{(n-1)}$ to the market. It is worth mentioning that there is no penalty on the first offer $b_i^{(1)}$.

$$\sigma_i^{(n)} = \begin{cases} \gamma \left| (b_i^{(n)} - b_i^{(n-1)}) / a_i \right| & n > 1 \\ 0 & \text{otherwise} \end{cases} \quad (56)$$

where, γ is the penalty coefficient of tentative offers.

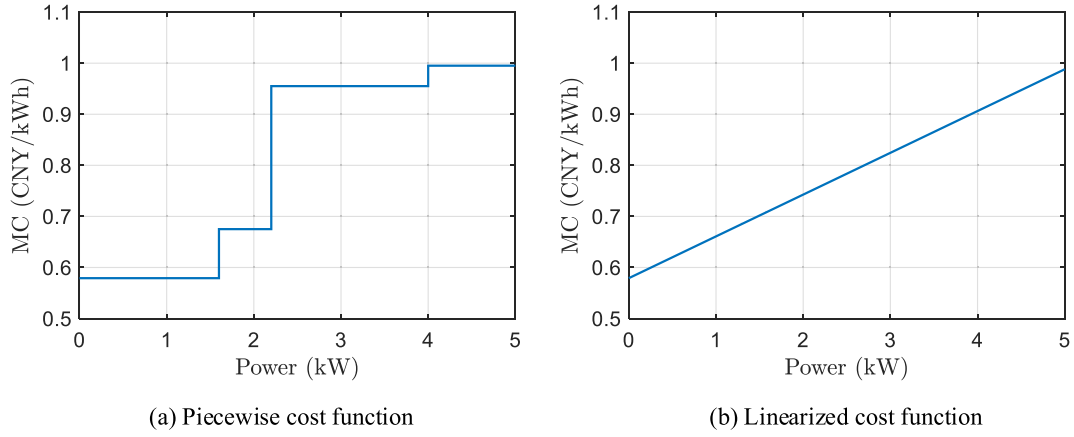


Fig. 2. Piecewise and linearized cost function of residential prosumer #1 ~ #45.

Table 3

Typical variable cost and fixed cost coefficients of prosumers during 12:00–12:15.

Parameters	Residential Prosumer (#1-#45)	Residential Prosumer (#46-#90)	Commercial Prosumer (#91-#96)	Commercial Prosumer (#97-#102)
c_i	0.0409	0.0336	0.0084	0.0113
d_i	0.5791	0.5287	1.0284	1.2342
a_i	24.4794	29.7599	119.0033	88.3095
b_i^*	-15.6390	-18.2638	-102.7653	-85.3451

In Section 3.2.2, we proved that the a_i of each prosumer would not affect the MCP as well as the trading quantity and profit of prosumers. Therefore, each individual prosumer and the aggregator will adjust their a_i as follows to turn $b_i^{(n)}$ into a constant, and thus avoid the penalty of tentative offers without affecting profit as follows:

$$a_i = \frac{1}{2c_i} \Rightarrow b_i^{(n)} = \frac{(1 - 2a_i c_i) c_{MCP}^{(n-1)} - d_i}{2c_i} = -\frac{d_i}{2c_i} \quad (57)$$

$$a_g = \frac{1}{2(c_g + \hat{\mu}_g/2a_g)} \Rightarrow b_g^{(n)} = -\frac{d_g}{2(c_g + \hat{\mu}_g/2a_g)} \quad (58)$$

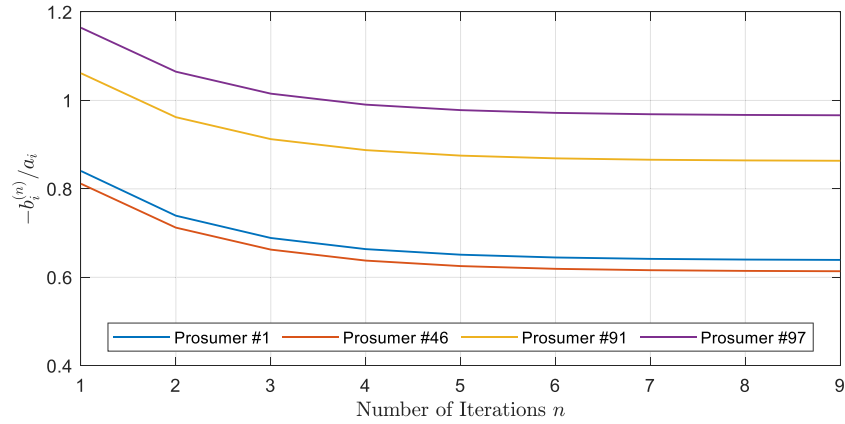


Fig. 3. Iterative pricing process of typical prosumers.

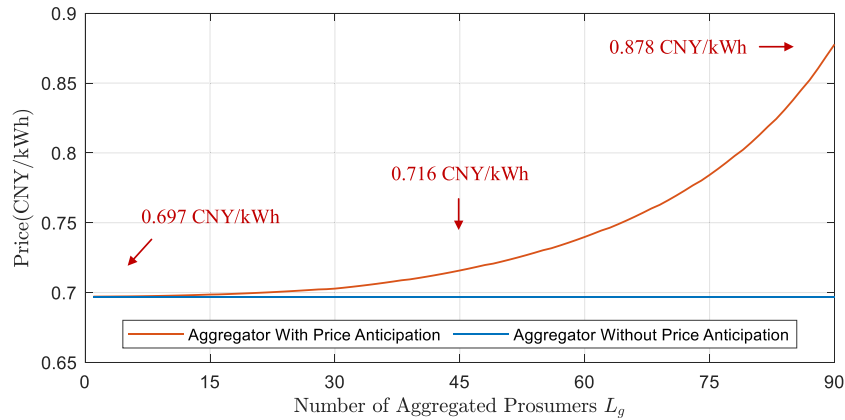


Fig. 4. Market clearing price during 12:00–12:15 with prosumer aggregation.

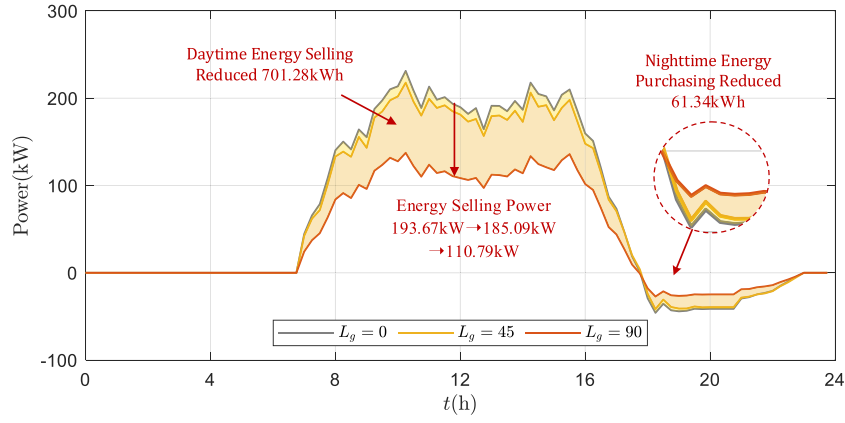


Fig. 5. Energy trading quantity between residential and commercial prosumers.

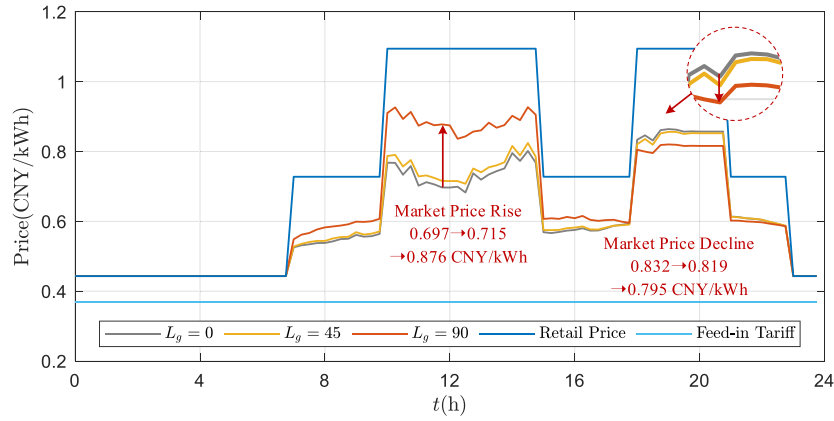


Fig. 6. Market clearing price of community microgrid with prosumer aggregation.

In Section 3.2.2, we also proved that the aggregator g might suffer losses when the price impact factor is overestimated. Therefore, when market power perception capability is prevented, the aggregator will adopt a conservative estimation of the price impact factor $\hat{\mu}_g < \mu_g^*$ in the trading strategy to avoid losses. Since a conservative $\hat{\mu}_g$ will lead to a weakened price anticipation ability of the aggregator, the impact of market power on the trading fairness and efficiency of the market will be restrained.

In addition to the above methods, the market power of aggregators can also be restrained in the following ways:

1) Improve the overall supply and demand elasticity of the market by

increasing the capacity of distributed energy storage and carrying out demand-side management. Then, the impact of aggregators' trading behaviors on market prices will be restrained.

2) During the peak period of energy production and consumption with relatively limited market elasticity, the P2PMO will restrict the trading behavior of aggregators by adjusting the price limit and formulating trading quotas.

3) The P2PMO will set the market access regulation for aggregators to avoid the emergence of strong market participants whose energy resource capacity accounts for more than 20% in the P2P energy trading market.

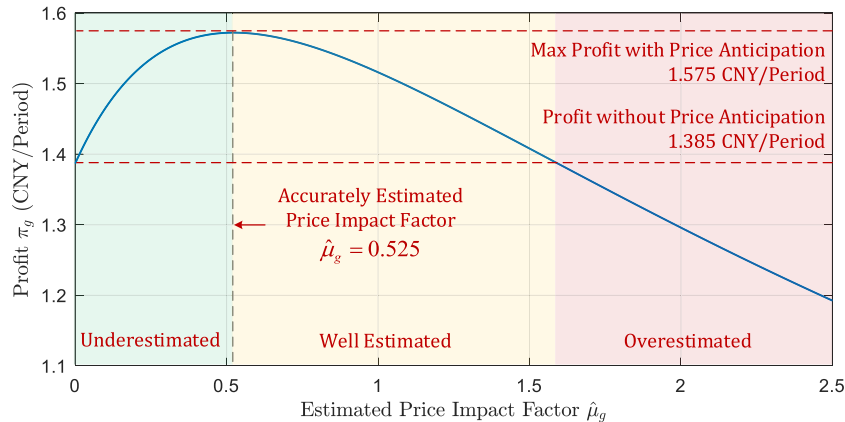


Fig. 7. Trading profit of the aggregator under different estimated price impact factor.

Table 4
Impact of prosumer aggregation on the trading profit of prosumers (CNY/Period).

Aggregation	Residential Prosumer Total Profit (#1-#45)	Residential Prosumer Total Profit (#46-#90)	Commercial Prosumer Total Profit (#91-#96)	Commercial Prosumer Total Profit (#97-#102)	Total Profit (#1-#102)
$L_g = 0$	1.387	2.373	3.601	7.018	14.379
$L_g = 45$	1.572	2.918	3.242	6.510	14.241

As a supplement to the methods proposed in this paper, the above aspects will be further explored in future studies.

4. Case study

A typical community microgrid system in [9] is taken as a test case to verify the proposed model and method. There are 90 residential prosumers (#1-#90) and 12 commercial prosumers (#91-#102) in the microgrid, and the typical energy resources configuration of each type of prosumer is shown in Table 1. Various types of prosumers are separately listed due to their differentiated and complementary energy production and consumption behavior. Each prosumer will participate in P2P energy trading to maximize its profit. Prosumers can also sell or purchase energy from the utility grid when the market cannot guarantee the energy balance. The retail price and feed-in tariff of the utility grid during each period are listed in Table 2 [34]. The P2P energy trading market simulation is performed with the method proposed in [9]. The mixed-integer linear programming (MILP) based mathematical model for P2P energy trading was solved with the CPLEX solver in MATLAB 2022b, and the total simulation time for the full-day market trading (0:00-24:00) was 6.8 h.

4.1. Results of market price formation

In the process of energy trading, prosumers will first determine the quantity of energy supply and demand as well as their marginal cost and utility. Then, generate the piecewise cost function accordingly with the supply and demand generation method proposed in [9], as shown in Fig. 2(a). Next, linearize the piecewise cost function into Eq.(10) with the least-squares method, as shown in Fig. 2(b). Take the trading period $t = 48$ (12:00-12:15) as an example. The c_i and d_i in the cost function of typical prosumers are listed in Table 3. In the trading period, prosumers iteratively submit the a_i and the optimized $b_i^{(n)}$ to P2PMO until the market reaches equilibrium, as shown in Table 3 and Fig. 3.

When prosumer aggregation is formed in the microgrid, the MCP may be affected by the market power of the aggregator g . Take the trading period $t = 48$ (12:00-12:15) as an example. When a different number of residential prosumers $L_g \in [0, 90]$ participate in the broker service of aggregator g , the MCP of the market is shown in Fig. 4.

Table 5
Impact of prosumer aggregation on overall benefit.

Evaluation Index	$L_g = 0$	$L_g = 45$	$L_g = 90$
Utility Grid Energy Exchange (kW)	1449.43	1569.57	2093.83
Utility Grid Energy Exchange Cost (CNY/d)	757.26	765.61	808.62
Microgrid Overall Economic Benefit (CNY/d)	355.81	343.72	285.99
Price of Anarchy	1.00	1.04	1.24
Renewable Energy Self-Consumption Ratio (%)	87.45%	85.61%	77.57%

Specifically, when the aggregator g performs price anticipation in the trading strategy, the market power of the aggregator leads to an increased MCP, which is beneficial for aggregated prosumers since they are selling energy. On the contrary, the MCP remains unchanged when the aggregator does not exploit the market power for profit. The above MCP results show that the substantial reason for the market equilibrium to deviate and the MCP to change is the market power abuse of the aggregator, which is in accordance with Remark 1.

Then, we further investigate the impact of market power on price formation from a longer time scale. When a specific number of residential prosumers are engaged in prosumer aggregation, the energy trading quantity and price between residential prosumers and commercial prosumers are shown in Fig. 5 and Fig. 6, respectively. Take daytime energy selling periods (7:00-18:00) as an example. When residential prosumers #1 ~ #45 are engaged in prosumer aggregation ($L_g = 45$), the aggregator reduces the energy supply and manipulates the market price to rise for profit, as shown in Fig. 6, which leads to reduced energy exchange between residential and commercial prosumers, as shown in Fig. 5. When more participants #1 ~ #90 are engaged in prosumer aggregation ($L_g = 90$), the market power of the aggregator will be more significant, which results in a further reduced energy supply and a higher price. Similarly, during the nighttime energy purchasing periods (18:00-23:00), the residential prosumers purchase less energy from commercial prosumers at a lower price. The above results show that the aggregator can manipulate the market price by adjusting the trading quantity, verifying Remark 2.

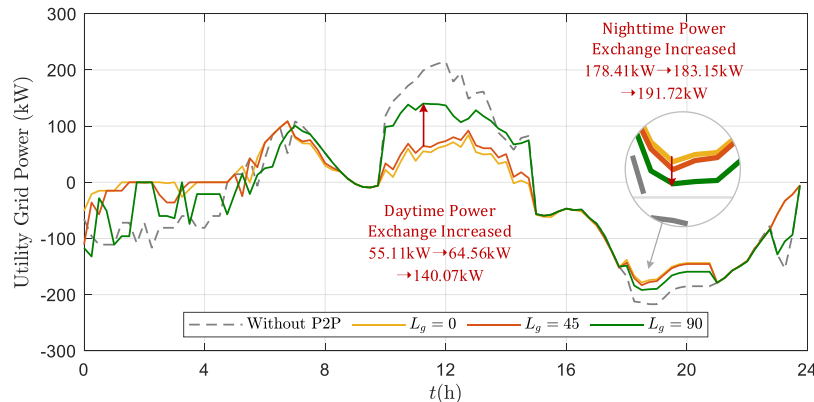


Fig. 8. Impact of prosumer aggregation on utility grid energy exchanging power.

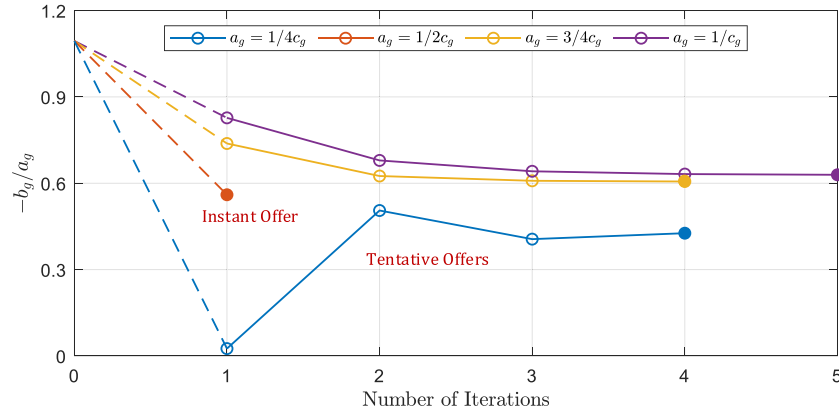


Fig. 9. Iterative bidding process of the aggregator.

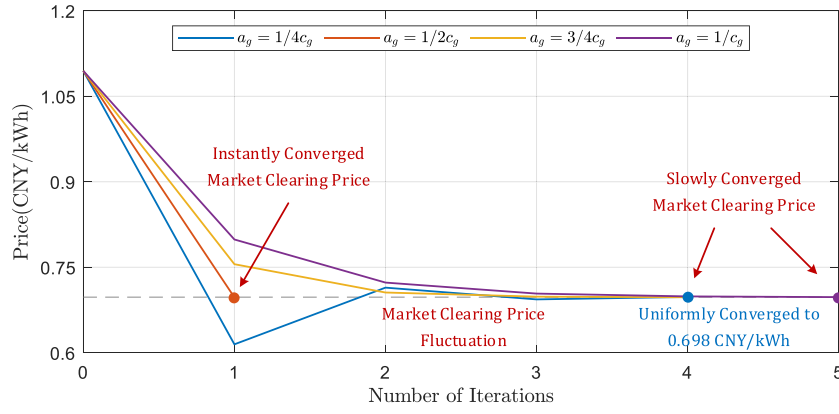


Fig. 10. Convergence process of the market clearing price.

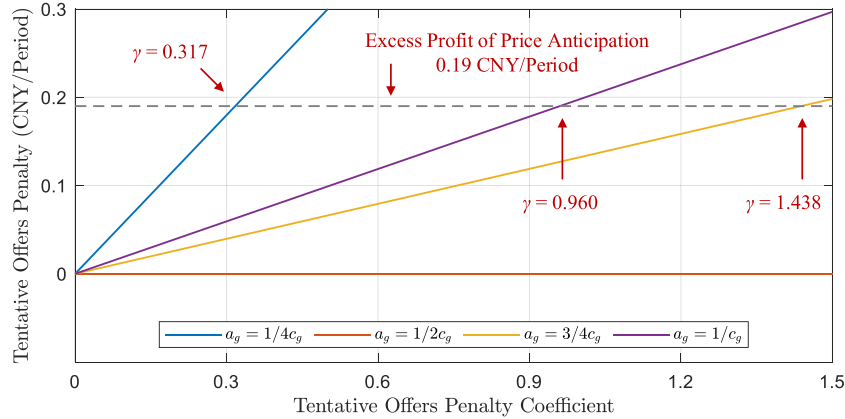


Fig. 11. Tentative offers penalty of the aggregator.

4.2. Results of market trading benefit

Take the trading period $t = 48$ (12:00–12:15) as an example. During this period, the residential prosumers are selling energy to commercial prosumers, as shown in Fig. 5. First, we investigate the case where residential prosumers #1 ~ #45 are involved in the aggregator g . When the aggregator performs price anticipation and the estimated price impact factor $\hat{\mu}_g$ of the aggregator g varies in the range of $[0, 2.5]$ times of μ_g , the resulting trading profit of aggregated prosumers is shown in Fig. 7. The result shows that the aggregator g can obtain higher profit when $\hat{\mu}_g$ in the trading strategy is well estimated. In particular, when $\hat{\mu}_g$

is accurately estimated with the method proposed in [23] as $\hat{\mu}_g = \mu_g = 0.525$, the aggregator can receive the maximum profit 1.575 CNY/Period. However, when $\hat{\mu}_g$ is overestimated (i.e., $\hat{\mu}_g > 3.985$ in this case), the aggregator will suffer losses (i.e., the profit is lower than 1.385 CNY/Period). The above results are in accordance with Proposition 3 and Remark 4.

Then, we further investigate the impact of market power on the profit of individual prosumers $i \notin \mathcal{D}_g$. The total trading profits of each type of prosumer before and after prosumer aggregation are shown in Table 4. It can be observed from the results that when residential prosumers #1 ~ #45 are engaged in prosumer aggregation $L_g = 45$, the

total energy selling profit of aggregated residential prosumers #1 ~ #45 increases, which verifies Remark 3. Meanwhile, the profit of energy-selling residential prosumers #46 ~ #90 also increases due to the 'hitchhiking effect' analyzed in Eq.(46)–(47). On the contrary, commercial prosumers #91 ~ #96 and #97 ~ #102 suffer energy purchasing benefit losses. Therefore, the market power of the aggregator results in unfair distribution of welfare among prosumers, verifying Remark 6.

Finally, we investigate the impact of prosumer aggregation on the overall benefit of the microgrid. The results in Fig. 5 show that the energy trading quantity between aggregated prosumers and individual prosumers is reduced when the aggregator performs price anticipation in the market for excess profit, verifying Proposition 4. As a result, more energy is exchanged between the microgrid and the utility grid during peak periods to compensate for the energy imbalance caused by the trading quantity curtailment of the aggregator. When a specific number of residential prosumers are engaged in prosumer aggregation $L_g \in \{45, 90\}$, the excessive energy exchange of the microgrid with the utility grid results in a weakened peak load shifting and renewable energy self-consumption capability of the market, as shown in Fig. 8 and Table 5. Meanwhile, the overall energy trading benefit of the microgrid also decreases due to excessive energy exchange (i.e., selling energy to the utility grid at a lower price of feed-in tariff while purchasing energy at a higher retail price), as shown in Table 5, verifying Proposition 5.

4.3. Results of market power restraint

To restrain the market power perception ability of the aggregator and further restrain the impact of market power on the fairness and efficiency of energy trading, the tentative offers penalty is imposed on the prosumers. At this point, individual prosumers will adopt Eq.(57) as the value strategy of a_i to turn $b_i^{(n)}$ into a constant, which will avoid the tentative offers penalty caused by the slow convergence of $b_i^{(n)}$ in the iterative market bidding.

On the contrary, the aggregator g can either adopt Eq.(57) as the value of a_i to submit an instant offer $b_g^{(1)}$ or stimulate price fluctuations by deliberately submit tentative offers $b_g^{(n)}$ to the market for market power perception. Take the trading period $t = 48$ (12:00–12:15) as an example. When the a_g varies in the range of $[1/4, 1]$ times of $1/c_g$, the iterative bidding process $b_g^{(n)}$ is shown in Fig. 9, and the convergence process of the MCP is shown in Fig. 10. The above results show that the aggregator can either take $a_g = 1/2c_g$ to facilitate an instant offer and avoid tentative offers penalty or take $a_g \neq 1/2c_g$ to stimulate market clearing price fluctuation and perform market power perception. Meanwhile, the results of Fig. 10 also prove that the MCP is not affected by the value of a_g , which is consistent with Remark 5.

In the particular case of $a_g = 1/c_g$, when the penalty coefficients taken as $\gamma > 0.163$, the tentative offers penalty imposed on the aggregator will be higher than the excess profit of market power perception and exploitation, as shown in Fig. 11. At this point, the prosumer will abort market power perception, taking Eq.(58) as the strategy of a_g to avoid the tentative offers penalty and adopting a conservative estimation of $\hat{\mu}_g$ to prevent losses of market power overestimation. As a result, the proposed method can effectively restrain the market power perception ability of the aggregator and further relieve the impact of market power on the benefit of the market.

5. Conclusion

A game-theoretic method for market power modeling and restraint of aggregated prosumers in Peer-to-Peer (P2P) energy trading is proposed in this paper. First, we established a framework for P2P energy trading market with the aggregator involved. Then, the competition between the aggregator and individual prosumers was modeled as a general non-

cooperative game, and we proved that a unique Nash equilibrium exists. Moreover, the impact of market power on price formation, trading profit, and overall benefit were analyzed, respectively. Finally, the tentative offers penalty was introduced in the market to restrain the market power of the aggregator. The conclusions drawn from the numerical results can be summarized as follows: 1) The aggregator can take advantage of its market power to manipulate the market price to increase its profit by 13.72% when the price impact factor is well perceived. 2) Market power may negatively affect the fair distribution of welfare among prosumers and the overall benefit of the microgrid. This leads to a Price of Anarchy (PoA) of 1.24 for the market, indicating that the efficiency of the microgrid system is reduced by 24% compared to the optimal level. 3) The tentative offers penalty proposed in this paper can effectively restrict the market power perception ability and further restrain the impact of market power on the benefit of the market. Future research will be carried out on the non-cooperative game between multiple aggregators, analyzing the impact of market power competition on the profit of individual prosumers and the overall benefit of the microgrid. Meanwhile, how to refine the market mechanism to guarantee the convergence of the bidding algorithm will also be studied.

CRedit authorship contribution statement

Zibo Wang: Writing – original draft, Software, Methodology, Conceptualization. **Lei Dong:** Writing – review & editing, Supervision. **Mengjie Shi:** Validation, Software. **Ji Qiao:** Visualization, Data curation. **Hongjie Jia:** Writing – review & editing, Supervision. **Yunfei Mu:** Writing – review & editing, Supervision. **Tianjiao Pu:** Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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References

- [1] Tushar W, Yuen C, Saha TK, Morstyn T, Chapman AC, Alam MJ, et al. Peer-to-peer energy systems for connected communities: a review of recent advances and emerging challenges. *Appl Energy* 2021;282:116131. <https://doi.org/10.1016/j.apenergy.2020.116131>.
- [2] Jin X, Wu Q, Jia H. Local flexibility markets: literature review on concepts, models and clearing methods. *Appl Energy* 2020;261:114387. <https://doi.org/10.1016/j.apenergy.2019.114387>.
- [3] Zafar R, Mahmood A, Razzaq S, Ali W, Naeem U, Shehzad K. Prosumer based energy management and sharing in smart grid. *Renew Sustain Energy Rev* 2018;82: 1675–84. <https://doi.org/10.1016/j.rser.2017.07.018>.
- [4] Nosratabadi SM, Hooshmand RA, Gholipour E. A comprehensive review on microgrid and virtual power plant concepts employed for distributed energy resources scheduling in power systems. *Renew Sustain Energy Rev* 2017;67: 341–63. <https://doi.org/10.1016/j.rser.2016.09.025>.
- [5] Görgülü H, Topçuoğlu Y, Yaldız A, Gökçek T, Ateş Y, Erdinç O. Peer-to-peer energy trading among smart homes considering responsive demand and interactive visual interface for monitoring. *Sustain Energy Grids Networks* 2022;29:100584. <https://doi.org/10.1016/j.segan.2021.100584>.
- [6] Morstyn T, Farrell N, Darby SJ, McCulloch MD. Using peer-to-peer energy-trading platforms to incentivize prosumers to form federated power plants. *Nat Energy* 2018;3(2):94–101. <https://doi.org/10.1038/s41560-017-0075-y>.
- [7] Zhang C, Wu J, Zhou Y, Cheng M, Long C. Peer-to-peer energy trading in a microgrid. *Appl Energy* 2018;220:1–2. <https://doi.org/10.1016/j.apenergy.2018.03.010>.

- [8] Gökçek T, Turan MT, Ateş Y, Arabal AY. A bi-level charging management approach for electric truck charging station considering power losses. *Turkish J Electric Eng Comput Sci* 2022;30(3):943–60. <https://doi.org/10.55730/1300-0632.3820>.
- [9] Wang Z, Yu X, Mu Y, Jia H. A distributed Peer-to-Peer energy transaction method for diversified prosumers in Urban Community Microgrid System. *Appl Energy* 2020;260:114327. <https://doi.org/10.1016/j.apenergy.2019.114327>.
- [10] Vergados DJ, Mamounakis I, Makris P, Varvarigos E. Prosumer clustering into virtual microgrids for cost reduction in renewable energy trading markets. *Sustain Energy Grids Networks* 2016;7:90–103. <https://doi.org/10.1016/j.segan.2016.06.002>.
- [11] Pei W, Du Y, Deng W, Sheng K, Xiao H, Qu H. Optimal bidding strategy and intramarket mechanism of microgrid aggregator in real-time balancing market. *IEEE Trans Industr Inform* 2016;12(2):587–96. <https://doi.org/10.1109/TII.2016.2522641>.
- [12] Cui S, Wang YW, Shi Y, Xiao JW. An efficient peer-to-peer energy-sharing framework for numerous community prosumers. *IEEE Trans Industr Inform* 2019;16(12):7402–12. <https://doi.org/10.1109/TII.2019.2960802>.
- [13] Iria J, Soares F, Matos M. Optimal bidding strategy for an aggregator of prosumers in energy and secondary reserve markets. *Appl Energy* 2019;238:1361–72. <https://doi.org/10.1016/j.apenergy.2019.01.191>.
- [14] Olivella-Rosell P, Lloret-Gallego P, Munné-Collado Í, Villafafila-Robles R, Sumper A, Ottessen SØ, et al. Local flexibility market design for aggregators providing multiple flexibility services at distribution network level. *Energies* 2018;11(4):822. <https://doi.org/10.3390/en11040822>.
- [15] Ghavidel S, Rajabi A, Jabbari Ghadi M, Azizivahed A, Li L, Zhang J. Hybrid power plant bidding strategy for voltage stability improvement, electricity market profit maximization, and congestion management. *IET Energy Systems Integration* 2021;3(2):130–41. <https://doi.org/10.1049/esi2.12008>.
- [16] Burger S, Chaves-Ávila JP, Batlle C, Pérez-Arriaga LJ. A review of the value of aggregators in electricity systems. *Renew Sustain Energy Rev* 2017;77:395–405. <https://doi.org/10.1016/j.rser.2017.04.014>.
- [17] Sheikhamadi P, Bahramara S. The participation of a renewable energy-based aggregator in real-time market: a Bi-level approach. *J Clean Prod* 2020;276:123149. <https://doi.org/10.1016/j.jclepro.2020.123149>.
- [18] Iria J, Soares F, Matos M. Optimal supply and demand bidding strategy for an aggregator of small prosumers. *Appl Energy* 2018;213:658–69. <https://doi.org/10.1016/j.apenergy.2017.09.002>.
- [19] Rashidizadeh-Kermani H, Vahedipour-Dahraie M, Shafie-khah M, Siano P. Optimal bidding of profit-seeking virtual associations of smart prosumers considering peer to peer energy sharing strategy. *Int J Electric Power Energy Syst* 2021;132:107175. <https://doi.org/10.1016/j.ijepes.2021.107175>.
- [20] Wang F, Ge X, Yang P, Li K, Mi Z, Siano P, et al. Day-ahead optimal bidding and scheduling strategies for DER aggregator considering responsive uncertainty under real-time pricing. *Energy* 2020;213:118765. <https://doi.org/10.1016/j.energy.2020.118765>.
- [21] Haghifam S, Dadashi M, Zare K, Seyedi H. Optimal operation of smart distribution networks in the presence of demand response aggregators and microgrid owners: a multi follower Bi-level approach. *Sustain Cities Soc* 2020;55:102033. <https://doi.org/10.1016/j.scs.2020.102033>.
- [22] Gazijahani FS, Salehi J. Game theory based profit maximization model for microgrid aggregators with presence of EDRP using information gap decision theory. *IEEE Syst J* 2018;13(2):1767–75. <https://doi.org/10.1109/JSYST.2018.2864578>.
- [23] Wang Z, Yu X, Mu Y, Jia H, Jiang Q, Wang X. Peer-to-peer energy trading strategy for energy balance service provider (EBSP) considering market elasticity in community microgrid. *Appl Energy* 2021;303:117596. <https://doi.org/10.1016/j.apenergy.2021.117596>.
- [24] Liu F, Bie Z, Jiang J, Wang K. Impact of microgrid aggregator on joint energy and reserve market based on pure strategy Nash equilibrium. *Energy Procedia* 2019;159:142–7. <https://doi.org/10.1016/j.egypro.2018.12.032>.
- [25] Chen Y, Zhao C, Low SH, Wierman A. An energy sharing mechanism considering network constraints and market power limitation. *IEEE Trans Smart Grid* 2023;14(2):1027–41. <https://doi.org/10.1109/TSG.2022.3198721>.
- [26] Faqiry MN, Das S. Double-sided energy auction in microgrid: equilibrium under price anticipation. *IEEE Access* 2016;4:3794–805. <https://doi.org/10.1109/ACCESS.2016.2591912>.
- [27] Ruhi NA, Dvijotham K, Chen N, Wierman A. Opportunities for price manipulation by aggregators in electricity markets. *IEEE Trans Smart Grid* 2017;9(6):5687–98. <https://doi.org/10.1109/TSG.2017.2694043>.
- [28] Khajeh H, Foroud AA, Firoozi H. Robust bidding strategies and scheduling of a price-maker microgrid aggregator participating in a pool-based electricity market. *IET Generat Transm Distribut* 2019;13(4):468–77. <https://doi.org/10.1049/iet-gtd.2018.5061>.
- [29] Contreras-Ocana JE, Ortega-Vazquez MA, Zhang B. Participation of an energy storage aggregator in electricity markets. *IEEE Trans Smart Grid* 2017;10(2):1171–83. <https://doi.org/10.1109/TSG.2017.2736787>.
- [30] Okajima Y, Hirata K, Gupta V. Economic impact and market power of strategic aggregators in energy. *Economically Enabled Energy Management: Interplay Between Control Engineering and Economics* 2020:153. https://doi.org/10.1007/978-981-15-3576-5_7.
- [31] Chen Y, Mei S, Zhou F, Low SH, Wei W, Liu F. An energy sharing game with generalized demand bidding: model and properties. *IEEE Trans Smart Grid* 2019;11(3):2055–66. <https://doi.org/10.1109/TSG.2019.2946602>.
- [32] Hobbs BF, Metzler CB, Pang JS. Strategic gaming analysis for electric power systems: an MPEC approach. *IEEE Trans Power Syst* 2000;15(2):638–45. <https://doi.org/10.1109/59.867153>.
- [33] Karthikeyan SP, Raglend LJ, Kothari DP. A review on market power in deregulated electricity market. *Int J Electric Power Energy Syst* 2013;48:139–47. <https://doi.org/10.1016/j.ijepes.2012.11.024>.
- [34] Kong X, Liu D, Wang C, Sun F, Li S. Optimal operation strategy for interconnected microgrids in market environment considering uncertainty. *Appl Energy* 2020;275:115336. <https://doi.org/10.1016/j.apenergy.2020.115336>.